

Connor Spinelli

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EDUCATION

Honors Student of Biomedical Engineering; Minor in Biomechanical Engineering

GPA: 3.62 – Dean's List (F'23, S'24, F'24, S'26) | University of Delaware (UD) | Newark, DE *Graduating May 2027*

Relevant Coursework: Differential Equations and Linear Algebra; Biostatistics; Clinical Biomechanics; Medical Device Design; Biological Transport Phenomena; Biomedical Instrumentation; Bioengineering Mechanics; Materials Science

WORK EXPERIENCE

Machine Shop Teaching Assistant | Design Studio, Spencer Lab, UD | Newark, DE

Aug 2025 - Present

- Work with a team of TA's to assist students in the design and manufacturing processes for their course projects, especially projects sponsored by external companies such as NASA, Merck, and Bloom Energy.
- Ensured laboratory safety and compliance for 100+ students while maintaining Design Studio facilities.
- Reducing part fabrication errors through hands-on process guidance with advanced metal forming machinery.

Undergraduate Researcher | Delaware INBRE | Newark, DE

Summer Internship June - August 2025

- Attended workshops to develop my skills in conducting research and career management as part of the Delaware IDeA Networks of Biomedical Research Excellence (INBRE) program.
- Designing EMG signal filters based on literature and comparing their effect on the significance of results under the supervision of an esteemed professor.
- Assisted in research, design, and manufacturing processes with post-graduate projects, including gaining proficiency in SolidWorks and MATLAB.

Assistant Researcher | Human Robotics Lab, UD | Newark, DE

August 2024 - October 2025

- Supported the development of neural stimulation robots and research into forearm muscle reflex responses (LLR).
- Designed the adjustable MRI DMSW Support Structure using SolidWorks and sourced MRI-compatible materials to meet strict scanner constraints.

PROJECTS ([Portfolio](#))

Wrist Support Structure for Catheter Insertion - Junior Design

Spring 2026

- Reimagined the current design for a device that can be mass-manufactured and used by medical professionals in the ICU, including wrist extension angle adjustability and an attachment for catheter securement

MRI Robot Support Structure

Spring 2025

- Engineered an adjustable, MRI-compatible support structure using non-ferrous materials to significantly minimize the difficulty in patient maneuvering during setup.

The Concentration of Digoxin Activity after Intravenous Administration in Three Compartments

Spring 2026

- Leveraged Claude AI iteratively throughout development to accelerate MATLAB implementation, debug stiff ODE solver logic, and pressure-test modeling assumptions, cutting simulation build time significantly
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LEADERSHIP

Vice President of Judicial Affairs & Expansion | Interfraternity Council | UD

May 2025 - Present

- Managed compliance and policy execution for 29 chapters (1800+ members), utilizing strong organizational and persuasive communication skills to drive accountability.

2nd Counselor, Vice President | Sigma Pi Iota-Beta Chapter | UD

November 2024 - November 2025

- Directed internal relations by managing 21 chair positions and their respective committees, streamlining organizational processes to ensure effective execution of biweekly events.
- Assist in conducting weekly chapter meetings by outlining goals, facilitating discussions, and increasing morale.

SKILLS

AutoCAD - Fusion 360, SOLIDWORKS, Python, Matlab, Microsoft Office, Image J, JMP Statistical Software, Soldering, Laser Cutting, Breadboard Circuits, Woodshop, Milling, CNC

CERTIFICATIONS

Matlab Onramp, Simulink Onramp, Signal Processing Onramp, Human Subjects Protections - Biomedical Focus, Responsible Conduct of Research, CITI Good Clinical Practice Course

PERSONAL INTERESTS

Skiing, Golfing, Backpacking/Hiking, Mixing music, Weightlifting, Traveling